

Challenges in Semiconductor Inspection

A comprehensive guide for AI Researchers

PRESENTED BY

Divya.R
Deepika.P

V.S.B Engineering College

Challenges in Semiconductor Inspection

Importance of Inspection

Semiconductor inspection ensures quality, reliability, and performance in electronic devices, impacting global technology.

Need for AI Solutions

AI-based restoration can enhance image quality, automate processes, and improve accuracy in semiconductor inspections.

Image Degradation Issues

Common issues include noise, blur, low contrast, and artifacts, complicating defect detection during inspections.



AI Restoration in Semiconductor Inspection

Importance of Inspection

Semiconductor inspection ensures product quality, impacting performance and reliability in electronics.

Challenges in Manual Inspection

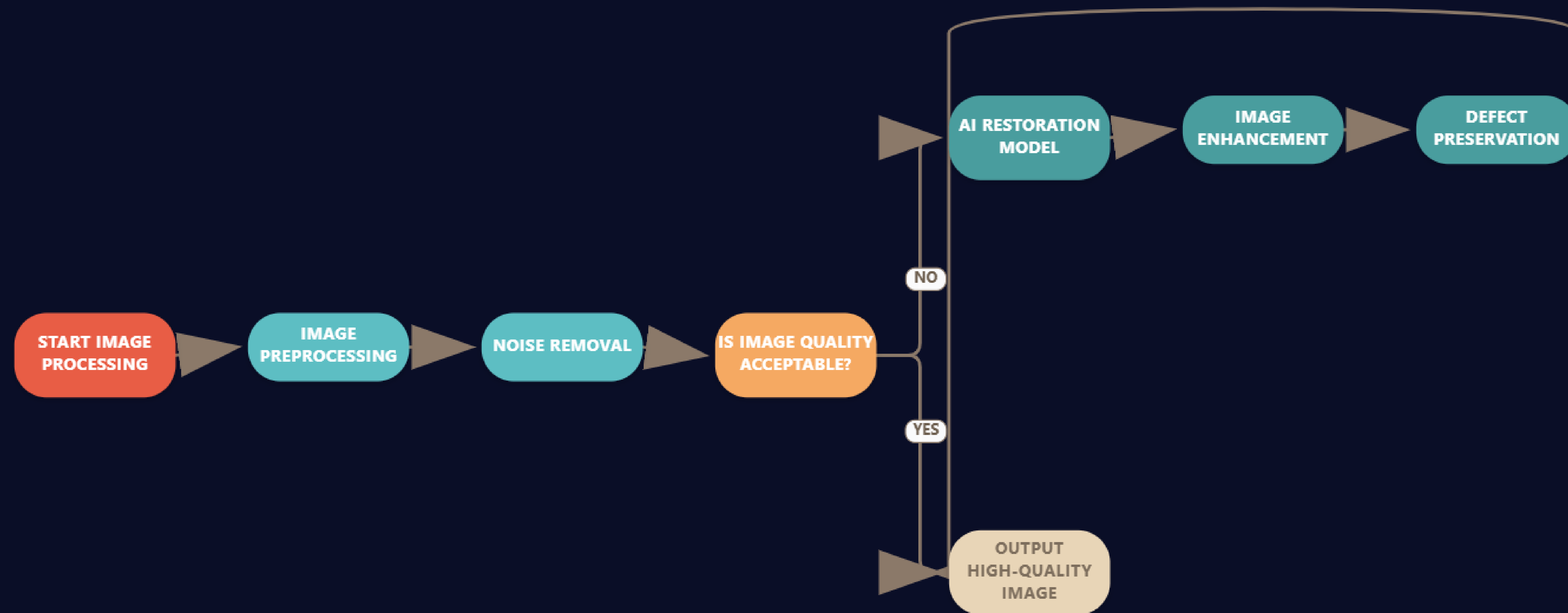
Manual inspection is time-consuming and prone to human error, necessitating automated solutions.

Image Degradation Issues

Common issues include noise, blur, and artifacts, complicating accurate defect detection.



AI Restoration Process Workflow



Technology Stack & System Architecture

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Programming Language

Python is chosen for its extensive libraries and community support in AI development.

Key Libraries

Utilizing OpenCV, TensorFlow, and PyTorch for image processing and deep learning tasks.

Framework & IDE

Flask for web deployment; VS Code for efficient coding and debugging.

Key Features & Expected Outcomes

AI Denoising

Utilizes advanced algorithms to effectively reduce noise, enhancing image clarity for better inspection.

Image Deblurring

Restores sharpness in images, allowing for precise defect identification and analysis during inspections.

Super Resolution

Enhances image resolution, enabling detection of minute defects that traditional methods may overlook.

Conclusion

The VisionChip AI project aims to revolutionize semiconductor inspection through advanced image restoration techniques. By leveraging AI, we enhance image quality, improve defect detection, and streamline manufacturing processes, paving the way for future innovations in smart manufacturing.